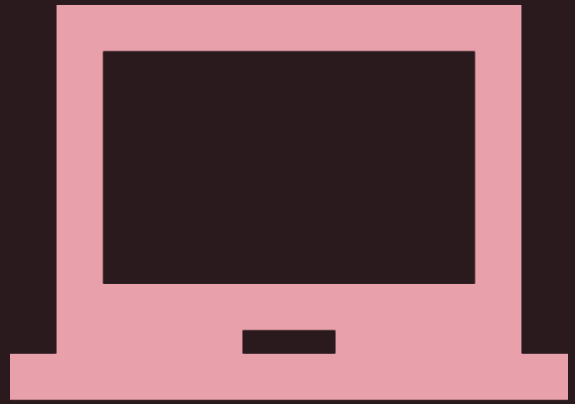
**Higher National Diploma in Network Engineering, Year 2**
**BSc (Hons) Ethical Hacking & Network Security, Year 2**

PRINCIPLES OF INFORMATION SECURITY

# Before Day 1: your laptop and the module



*Before Day 1*

Running case: Taxila College, a fictional college

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# What you will be able to do

**LO1** Explain the fundamentals of information security.

**LO2** Discuss the key characteristics of cryptography and cryptographic systems.

**LO3** Analyze the principles of encryption, decryption, hashing, and data integrity.

**LO4** Outline the process of ensuring digital security.

**LO5** Describe the fundamentals of threats and countermeasures.

**LO6** Explain the fundamentals and process of business continuity planning.

CONCEPT 1

# The shape of the ten days

*Ten days, each one built on the days before it.*

# Nine teaching days, then a day for the group and the exam

- Each teaching day has a **morning** of ideas and an **afternoon** lab or activity
- Every idea is shown on a small demo **before** it is given its name
- Each day builds only on the days before it
- Day 10 holds the group exercise and exam preparation

T TAXILA

You join Taxila College's information technology office as a trainee security analyst. Each teaching day is one day of your placement.

# The ten days

## Days 1 to 5

- Day 1: what we protect, and from what
- Day 2: secrecy with keys
- Day 3: hashing and integrity
- Day 4: signatures and certificates
- Day 5: who you are, what you may do

## Days 6 to 10

- Day 6: web application flaws and fixes
- Day 7: email and databases
- Day 8: networks, threats and defences
- Day 9: attacks, risk and continuity
- Day 10: group exercise and exam preparation

**If you miss a day, read its notes before the next one.**

# Each day stands on the one before it

- Morning: watch the idea work in a demo, then name it
- Afternoon: use it on your own laptop
- Between days: notes, quick checks, plays and one practice question

YOU ALREADY KNOW THIS

A building goes up one floor at a time. You cannot stand on a floor before the one below it is built.

MAPPING

**each floor =** one teaching day · **the floor below =** the earlier days · **standing on an unbuilt floor =** using an idea before the idea it needs

CONCEPT 2

# Two ways to run every lab

*Every lab works on every laptop.*

# Floor tier and full tier

## Floor tier

### What you need

A current web browser. Every laptop can do this.

### What you use

- The portal's demos and the lab's files
- CyberChef, a tool that runs in the browser

## Full tier

### What you need

8 GB of memory, 40 GB free, an Intel or AMD processor.

### What you use

- A Kali Linux virtual machine in VirtualBox
- OpenSSL, Wireshark, Lynis, iptables, MariaDB, Python

**Every mark can be earned at the floor tier.**

# The full tier is closer to real work, not worth more marks

YOU ALREADY KNOW THIS

You can cook every dish in a recipe book on a single hob. A full cooker is faster for some dishes, but it is not needed.

MAPPING

**the single hob =** the floor tier, enough for every lab · **the full cooker =** the full tier, faster but not required

× COMMON MISTAKE

*The floor tier earns fewer marks.*

**Every mark can be earned in the browser. The virtual machine is optional.**

# Before Day 1: set up and check

1. Read the getting-started guide on the portal
2. Set up the floor tier, and keep offline copies of the portal and CyberChef
3. If your laptop can run it, set up the full tier and take the `prelude-ready` snapshot
4. Open the environment check page and read its result
5. Bring the laptop, charged, to Day 1

**The check page should say Floor tier ready.**

CONCEPT 3

# Working within the law

*You learn attacks from the defender's seat.*

# You verify, configure, fix, detect and record

- Attacks are explained, and shown the way a defender sees them
- You never run an attack tool or type an attack input
- Labs use only your own machine, the provided files and applications, and the portal's demos
- Every lab opens its rules of engagement the same way: no tool in this lab can affect anyone else's system

**T** TAXILA

The trainee analyst works only inside the office's own practice set-up, never on the college's real systems.

# Permission is the line the law draws

- Sri Lanka's Computer Crime Act, No. 24 of 2007, makes access without lawful authority an offence
- Testing a system needs written permission from its owner
- Good intentions are not permission; doing no damage is not permission

× COMMON MISTAKE

*Trying a technique on a real website is harmless if nothing breaks.*

**Without written permission it is an offence, whatever happens.**

# Stay on the practice track

**Not sure whether something is allowed? Ask before you try it.**

YOU ALREADY KNOW THIS

A learner driver practises on a closed track. The same driving on a public road, without a licence, breaks the law even if nobody is hurt.

MAPPING

**the practice track =** your own machine and the provided material · **the public road without a licence =** someone else's system without permission

CONCEPT 4

# How your coursework stays yours

*Values only you produce, a sketch only you drew.*

# Five evidence packs, after Days 2 to 6

**Part 1**
## 8 marks

### Your personal values

Each lab works from your index number, written without spaces or hyphens.

*The lecturer works them out again.*

**Part 2**
## 5 marks

### A sketch, by hand

Draw the day's idea on paper, label its parts, and photograph it.

*Labels are marked, not neatness.*

**Part 3**
## 3 marks

### A short explanation

Answer the lab's one "why" question in your own words.

*At most 120 words.*

Completeness adds 4 marks. Each pack is due **before the next teaching day**.

# Your values are yours alone

YOU ALREADY KNOW THIS

A photograph of your own handwriting is yours. Someone else's handwriting in it would look wrong at once.

MAPPING

**your handwriting =** values from your own index number · **someone else's handwriting =** copied values, which are wrong for your index

T TAXILA

Every pack tells Taxila's story, but the numbers in it are yours alone.

# The group tabletop exercise

- Groups of four form on Day 5, and each member takes one role
- Across Days 6 to 9 your group builds a risk register and a controls map
- On Day 10 your group handles a live incident at Taxila and logs each action
- Each member answers one question about a role that was not their own

# Your work, your understanding

- Marks reward understanding that you show and explain
- A partly correct answer, explained well, can earn more than a perfect one you cannot explain
- You may use artificial intelligence (AI) tools to learn; your programme's rules on generated text apply
- You may be asked to explain any part of your coursework in a short viva

**Use AI, understand what it gives you, own what you submit.**

# Common mistakes

× COMMON MISTAKE

*The floor tier earns fewer marks.*

**Every mark can be earned in the browser; the virtual machine is optional.**

× COMMON MISTAKE

*Trying a security technique on a real website is harmless if nothing breaks.*

**Without written permission it is an offence. Work only on your own machine and the provided material.**

# Before Day 1

| | |
|---|---|
| READ | **The getting-started guide**<br>How the days work, how to set up, and the rules of engagement. |
| TASK | **Set up your laptop**<br>Floor tier first, full tier if your laptop can run it.<br>*Then open the environment check page.* |
| SELF-CHECK | **The environment check page**<br>It should say Floor tier ready, online and offline. |

# Questions & Discussion

*Next: Day 1, what we protect, and from what.*



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Practise only on your own virtual machine, the module's provided files and application, and the portal's demos. Testing a system you do not own, without written permission, is an offence under Sri Lanka's Computer Crime Act, No. 24 of 2007.

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